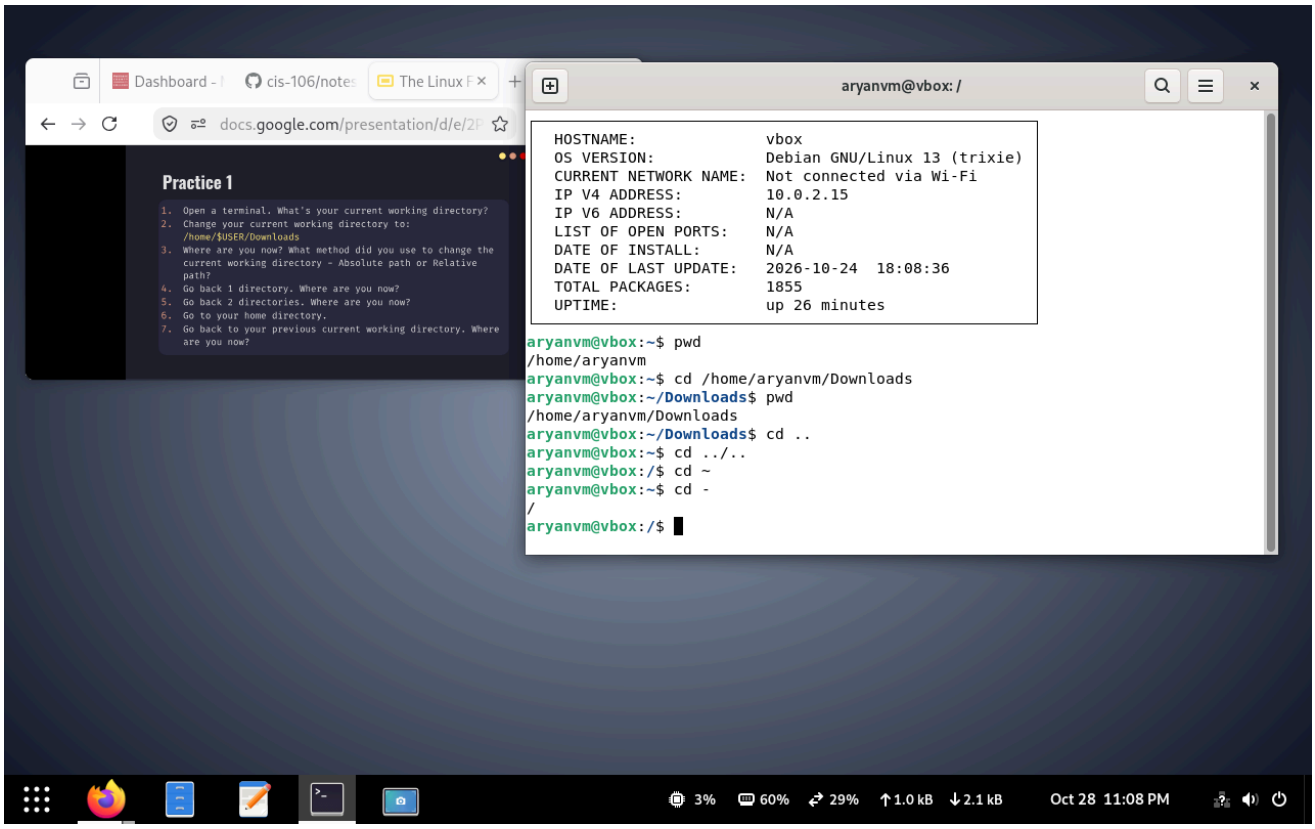
week report 5

completed work for week 3

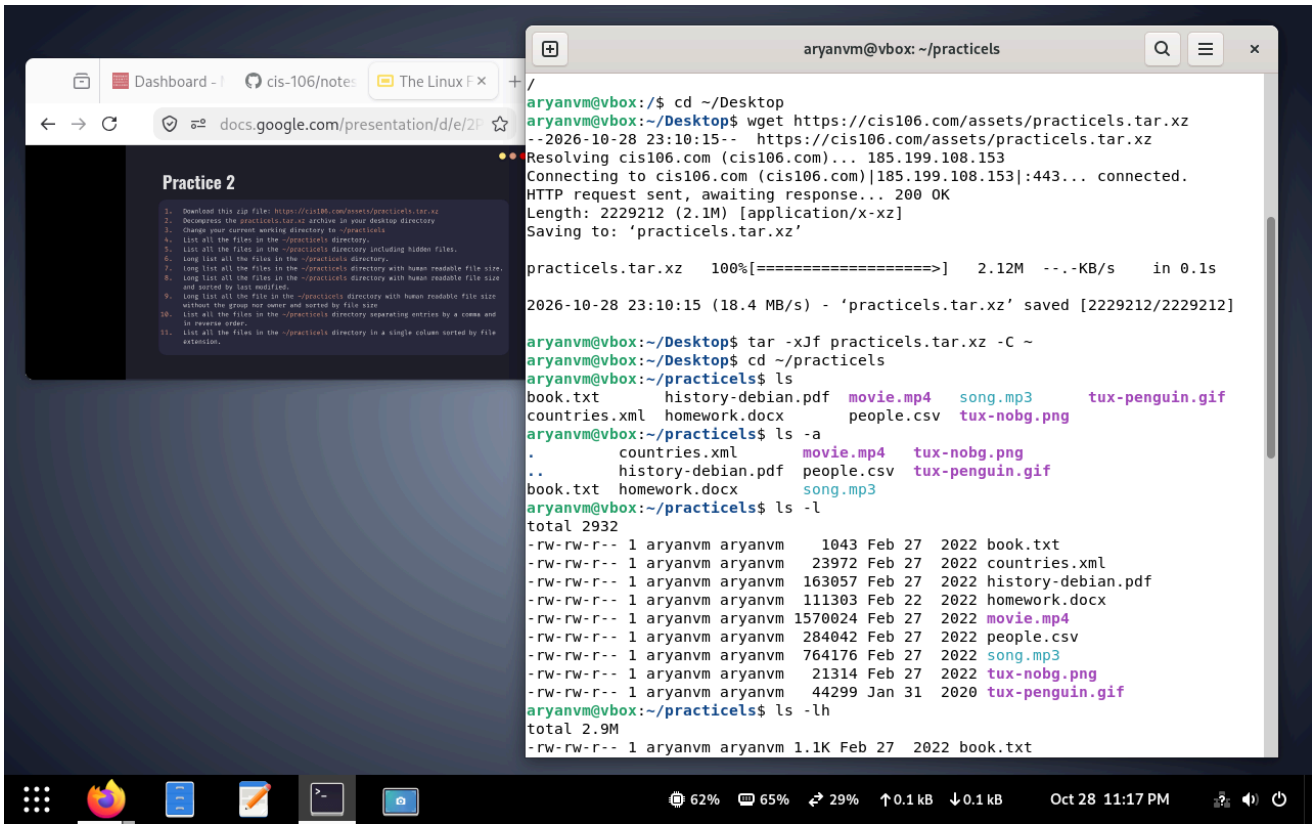
- lab 5
- notes 5

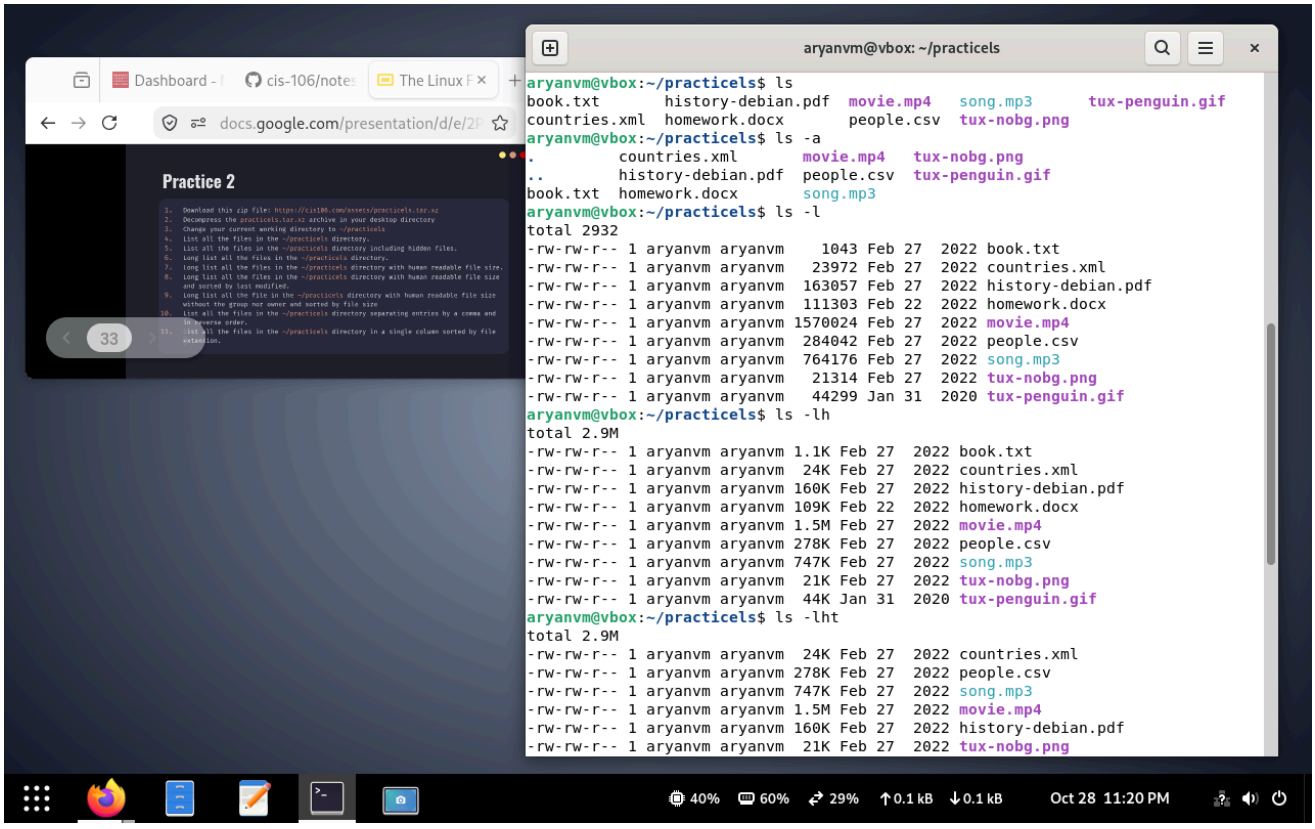
Practice screenshots :

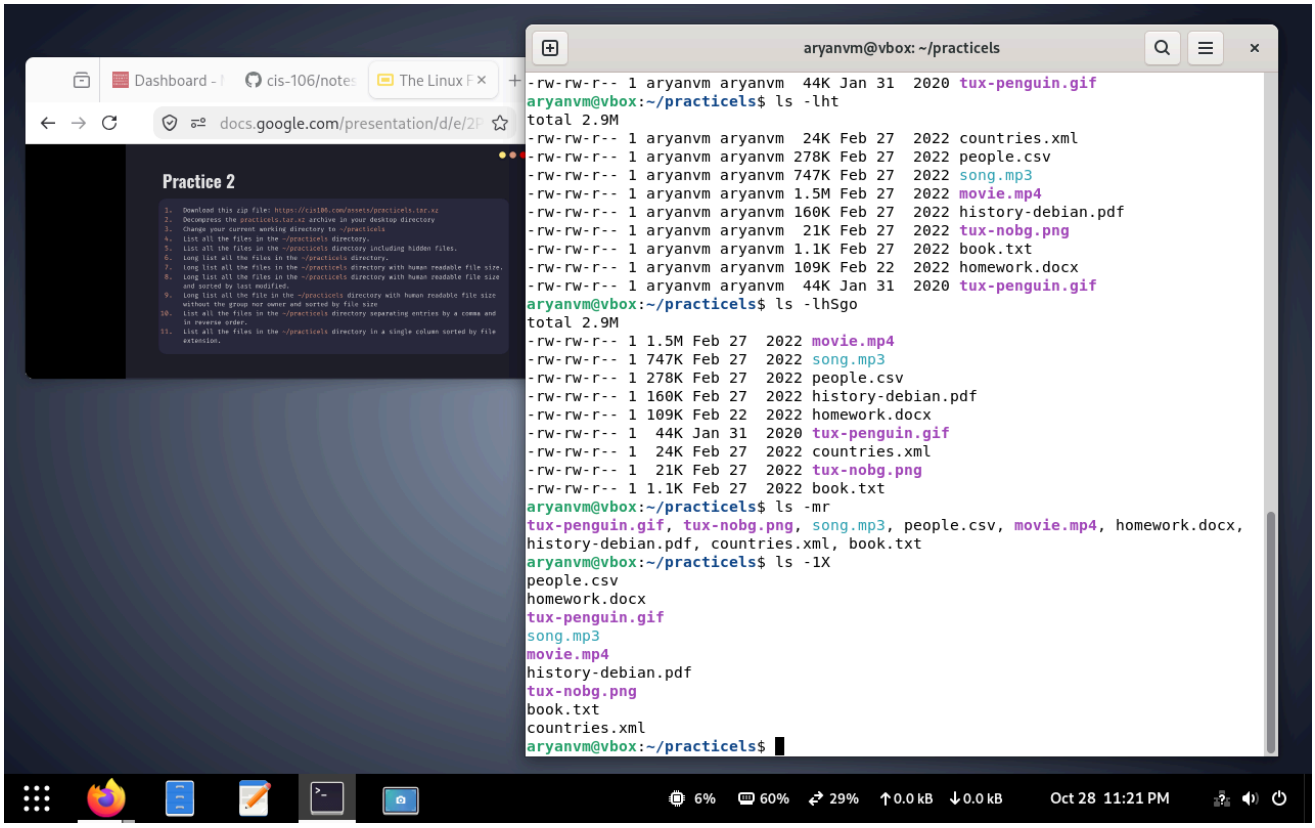
Practice 1



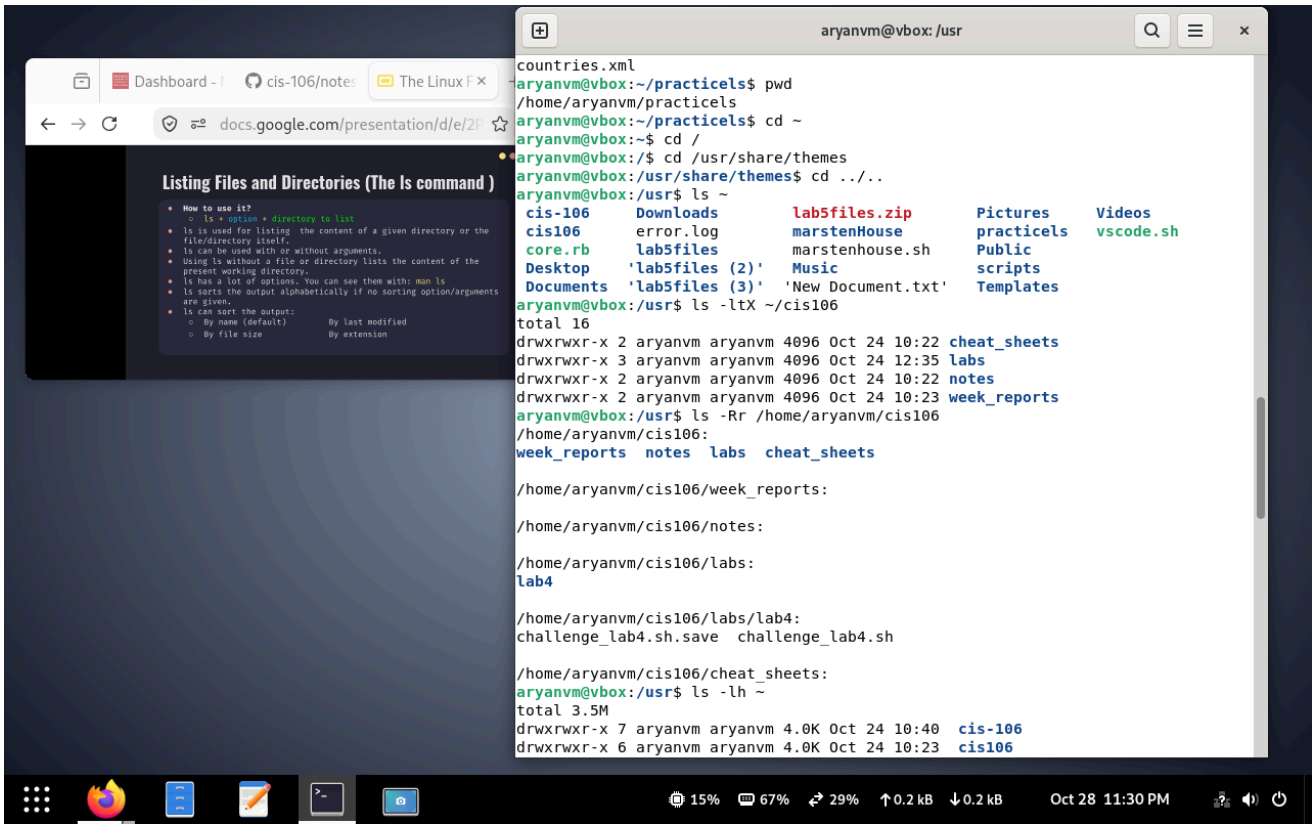
Practice 2

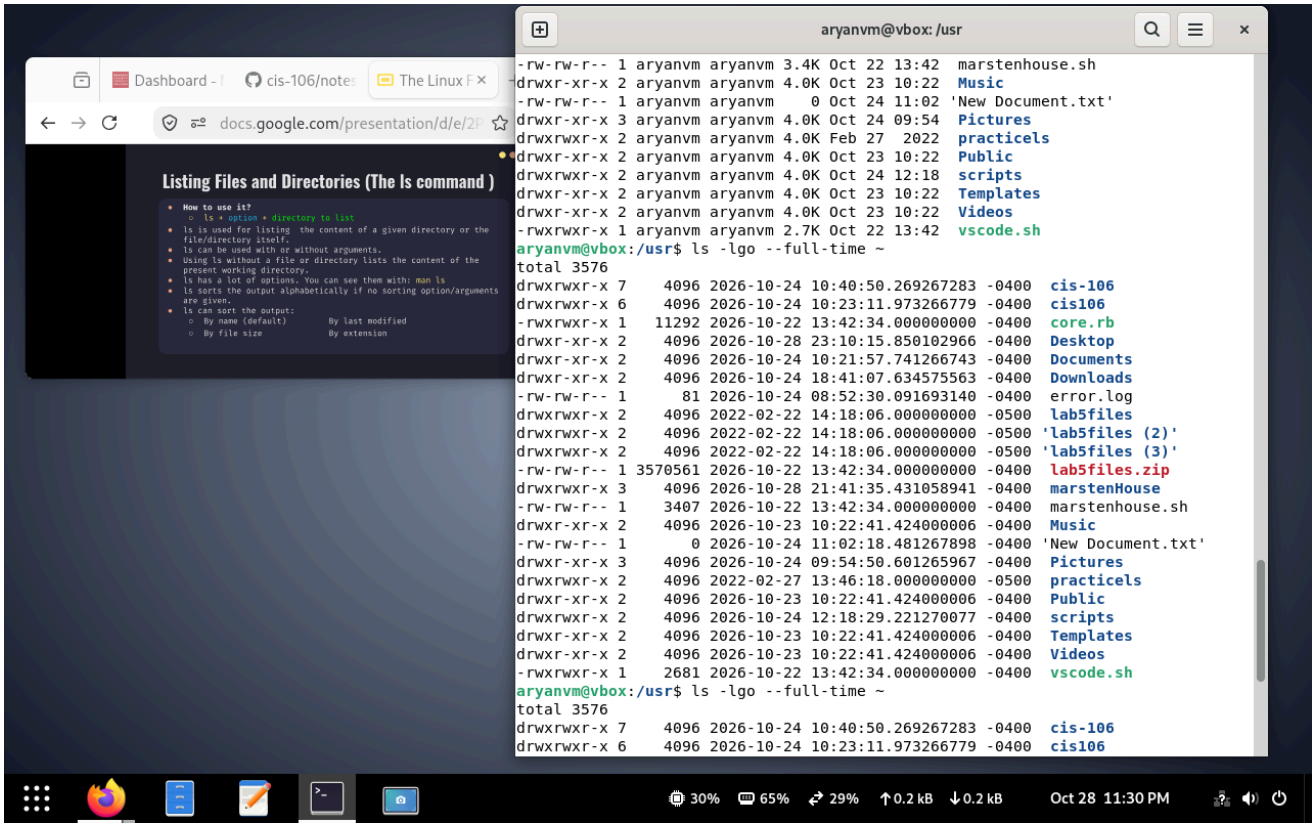

Practice 2
aryanvm@vbox: ~/practicels
/
aryanvm@vbox:/$ cd ~/Desktop
aryanvm@vbox:~/Desktop$ wget https://cis106.com/assets/practicels.tar.xz
--2026-10-28 23:10:15-- https://cis106.com/assets/practicels.tar.xz
Resolving cis106.com (cis106.com)... 185.199.108.153
Connecting to cis106.com (cis106.com)|185.199.108.153|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 2229212 (2.1M) [application/x-xz]
Saving to: 'practicels.tar.xz'
practicels.tar.xz 100%[==========>] 2.12M --.-KB/s in 0.1s
2026-10-28 23:10:15 (18.4 MB/s) - 'practicels.tar.xz' saved [2229212/2229212]
aryanvm@vbox:~/Desktop$ tar -xJf practicels.tar.xz -C ~
aryanvm@vbox:~/Desktop$ cd ~/practicels
aryanvm@vbox:~/practicels$ ls
book.txt history-debian.pdf movie.mp4 song.mp3 tux-penguin.gif
countries.xml homework.docx people.csv tux-nobg.png
aryanvm@vbox:~/practicels$ ls -a
. countries.xml movie.mp4 tux-nobg.png
.. history-debian.pdf people.csv tux-penguin.gif
book.txt homework.docx song.mp3
aryanvm@vbox:~/practicels$ ls -l
total 2932
-rw-rw-r-- 1 aryanvm aryanvm 1043 Feb 27 2022 book.txt
-rw-rw-r-- 1 aryanvm aryanvm 23972 Feb 27 2022 countries.xml
-rw-rw-r-- 1 aryanvm aryanvm 163057 Feb 27 2022 history-debian.pdf
-rw-rw-r-- 1 aryanvm aryanvm 111303 Feb 22 2022 homework.docx
-rw-rw-r-- 1 aryanvm aryanvm 1570024 Feb 27 2022 movie.mp4
-rw-rw-r-- 1 aryanvm aryanvm 284042 Feb 27 2022 people.csv
-rw-rw-r-- 1 aryanvm aryanvm 764176 Feb 27 2022 song.mp3
-rw-rw-r-- 1 aryanvm aryanvm 21314 Feb 27 2022 tux-nobg.png
-rw-rw-r-- 1 aryanvm aryanvm 44299 Jan 31 2020 tux-penguin.gif
aryanvm@vbox:~/practicels$ ls -lh
total 2.9M
-rw-rw-r-- 1 aryanvm aryanvm 1.1K Feb 27 2022 book.txt
62% 65% 29% ↑0.1 kB ↓0.1 kB Oct 28 11:17 PM


Practice 2
aryanvm@vbox: ~/practicels
aryanvm@vbox:~/practicels$ ls
book.txt history-debian.pdf movie.mp4 song.mp3 tux-penguin.gif
countries.xml homework.docx people.csv tux-nobg.png
aryanvm@vbox:~/practicels$ ls -a
. countries.xml movie.mp4 tux-nobg.png
.. history-debian.pdf people.csv tux-penguin.gif
book.txt homework.docx song.mp3
aryanvm@vbox:~/practicels$ ls -l
total 2932
-rw-rw-r-- 1 aryanvm aryanvm 1043 Feb 27 2022 book.txt
-rw-rw-r-- 1 aryanvm aryanvm 23972 Feb 27 2022 countries.xml
-rw-rw-r-- 1 aryanvm aryanvm 163057 Feb 27 2022 history-debian.pdf
-rw-rw-r-- 1 aryanvm aryanvm 111303 Feb 22 2022 homework.docx
-rw-rw-r-- 1 aryanvm aryanvm 1570024 Feb 27 2022 movie.mp4
-rw-rw-r-- 1 aryanvm aryanvm 284042 Feb 27 2022 people.csv
-rw-rw-r-- 1 aryanvm aryanvm 764176 Feb 27 2022 song.mp3
-rw-rw-r-- 1 aryanvm aryanvm 21314 Feb 27 2022 tux-nobg.png
-rw-rw-r-- 1 aryanvm aryanvm 44299 Jan 31 2020 tux-penguin.gif
aryanvm@vbox:~/practicels$ ls -lh
total 2.9M
-rw-rw-r-- 1 aryanvm aryanvm 1.1K Feb 27 2022 book.txt
-rw-rw-r-- 1 aryanvm aryanvm 24K Feb 27 2022 countries.xml
-rw-rw-r-- 1 aryanvm aryanvm 160K Feb 27 2022 history-debian.pdf
-rw-rw-r-- 1 aryanvm aryanvm 109K Feb 22 2022 homework.docx
-rw-rw-r-- 1 aryanvm aryanvm 1.5M Feb 27 2022 movie.mp4
-rw-rw-r-- 1 aryanvm aryanvm 278K Feb 27 2022 people.csv
-rw-rw-r-- 1 aryanvm aryanvm 747K Feb 27 2022 song.mp3
-rw-rw-r-- 1 aryanvm aryanvm 21K Feb 27 2022 tux-nobg.png
-rw-rw-r-- 1 aryanvm aryanvm 44K Jan 31 2020 tux-penguin.gif
aryanvm@vbox:~/practicels$ ls -lht
total 2.9M
-rw-rw-r-- 1 aryanvm aryanvm 24K Feb 27 2022 countries.xml
-rw-rw-r-- 1 aryanvm aryanvm 278K Feb 27 2022 people.csv
-rw-rw-r-- 1 aryanvm aryanvm 747K Feb 27 2022 song.mp3
-rw-rw-r-- 1 aryanvm aryanvm 1.5M Feb 27 2022 movie.mp4
-rw-rw-r-- 1 aryanvm aryanvm 160K Feb 27 2022 history-debian.pdf
-rw-rw-r-- 1 aryanvm aryanvm 21K Feb 27 2022 tux-nobg.png
40% 60% 29% ↑0.1 kB ↓0.1 kB Oct 28 11:20 PM

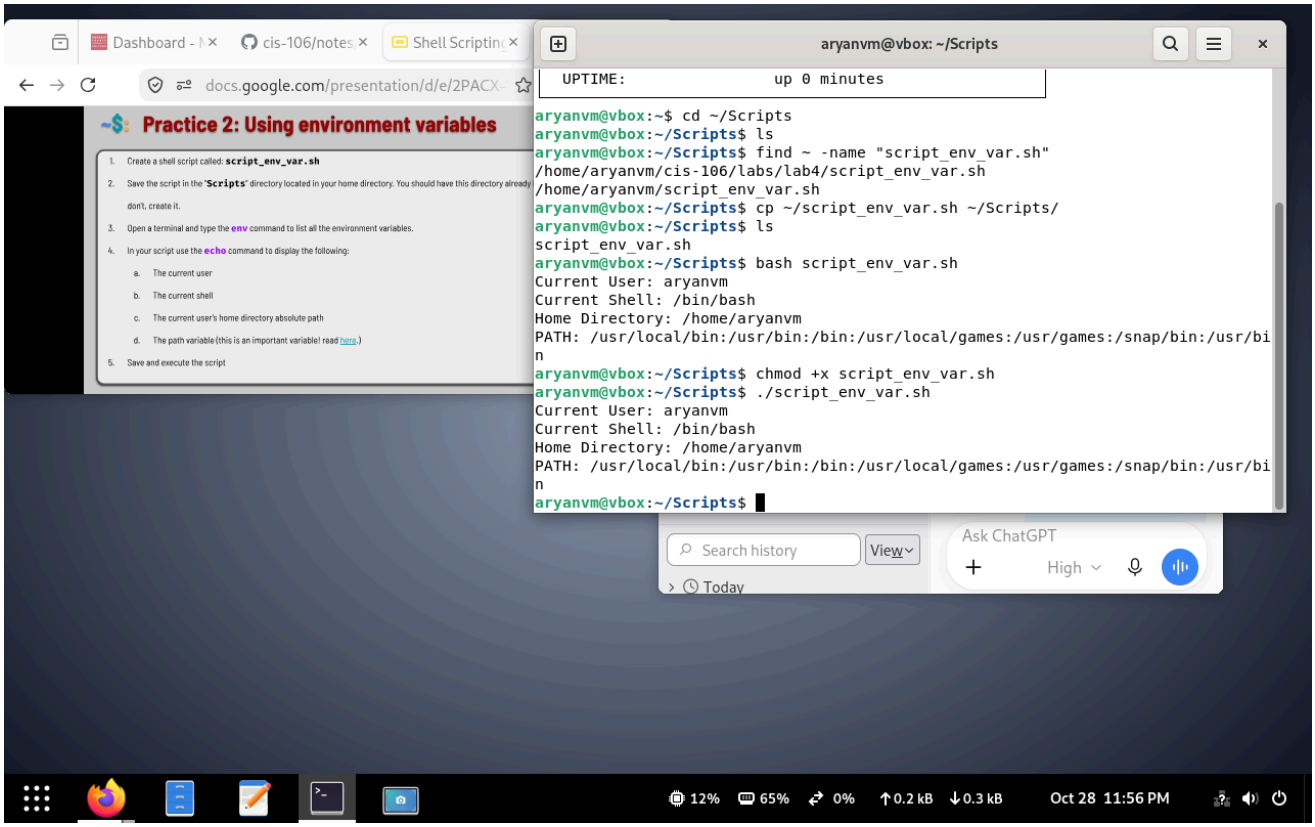

Dashboard - I
cis-106/notes
The Linux F
docs.google.com/presentation/d/e/2I
Practice 2
1. Download this zip file: https://cis106.com/assets/practicels.tar.xz
2. Decompress the practicels.tar.xz archive in your desktop directory
3. Change your current working directory to ~/practicels
4. List all the files in the ~/practicels directory.
5. List all the files in the ~/practicels directory including hidden files.
6. Long list all the files in the ~/practicels directory.
7. Long list all the files in the ~/practicels directory with human readable file size.
8. Long list all the files in the ~/practicels directory with human readable file size and sorted by last modified.
9. Long list all the file in the ~/practicels directory with human readable file size without the group nor owner and sorted by file size
10. List all the files in the ~/practicels directory separating entries by a comma and in reverse order.
11. List all the files in the ~/practicels directory in a single column sorted by file extension.
aryanvm@vbox: ~/practicels
-rw-rw-r-- 1 aryanvm aryanvm 44K Jan 31 2020 tux-penguin.gif
aryanvm@vbox:~/practicels$ ls -lht
total 2.9M
-rw-rw-r-- 1 aryanvm aryanvm 24K Feb 27 2022 countries.xml
-rw-rw-r-- 1 aryanvm aryanvm 278K Feb 27 2022 people.csv
-rw-rw-r-- 1 aryanvm aryanvm 747K Feb 27 2022 song.mp3
-rw-rw-r-- 1 aryanvm aryanvm 1.5M Feb 27 2022 movie.mp4
-rw-rw-r-- 1 aryanvm aryanvm 160K Feb 27 2022 history-debian.pdf
-rw-rw-r-- 1 aryanvm aryanvm 21K Feb 27 2022 tux-nobg.png
-rw-rw-r-- 1 aryanvm aryanvm 1.1K Feb 27 2022 book.txt
-rw-rw-r-- 1 aryanvm aryanvm 109K Feb 22 2022 homework.docx
-rw-rw-r-- 1 aryanvm aryanvm 44K Jan 31 2020 tux-penguin.gif
aryanvm@vbox:~/practicels$ ls -lhSgo
total 2.9M
-rw-rw-r-- 1 1.5M Feb 27 2022 movie.mp4
-rw-rw-r-- 1 747K Feb 27 2022 song.mp3
-rw-rw-r-- 1 278K Feb 27 2022 people.csv
-rw-rw-r-- 1 160K Feb 27 2022 history-debian.pdf
-rw-rw-r-- 1 109K Feb 22 2022 homework.docx
-rw-rw-r-- 1 44K Jan 31 2020 tux-penguin.gif
-rw-rw-r-- 1 24K Feb 27 2022 countries.xml
-rw-rw-r-- 1 21K Feb 27 2022 tux-nobg.png
-rw-rw-r-- 1 1.1K Feb 27 2022 book.txt
aryanvm@vbox:~/practicels$ ls -mr
tux-penguin.gif, tux-nobg.png, song.mp3, people.csv, movie.mp4, homework.docx,
history-debian.pdf, countries.xml, book.txt
aryanvm@vbox:~/practicels$ ls -1X
people.csv
homework.docx
tux-penguin.gif
song.mp3
movie.mp4
history-debian.pdf
tux-nobg.png
book.txt
countries.xml
aryanvm@vbox:~/practicels$
6%
60%
29%
0.0 kB
0.0 kB
Oct 28 11:21 PM

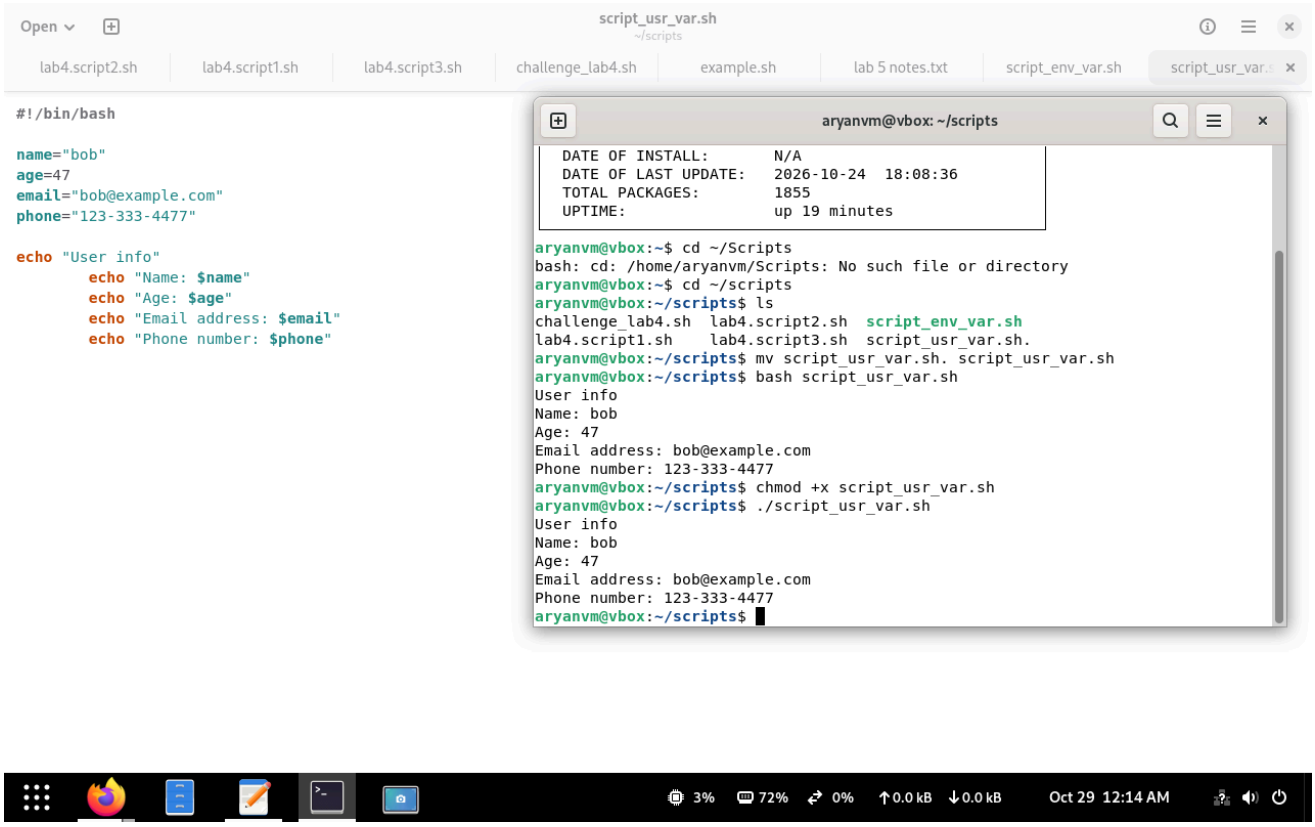
Practice 3


Dashboard - I
cis-106/notes
The Linux F
docs.google.com/presentation/d/e/2F
Listing Files and Directories (The ls command)
How to use it?
ls + option + directory to list
ls is used for listing the content of a given directory or the file/directory itself.
ls can be used with or without arguments.
Using ls without a file or directory lists the content of the present working directory.
ls has a lot of options. You can see them with: man ls
ls sorts the output alphabetically if no sorting option/arguments are given.
ls can sort the output:
By name (default)
By last modified
By file size
By extension
aryanvm@vbox: /usr
countries.xml
aryanvm@vbox:~/practicels$ pwd
/home/aryanvm/practicels
aryanvm@vbox:~/practicels$ cd ~
aryanvm@vbox:~$ cd /
aryanvm@vbox:/$ cd /usr/share/themes
aryanvm@vbox:/usr/share/themes$ cd ../..
aryanvm@vbox:/usr$ ls ~
cis-106 Downloads lab5files.zip Pictures Videos
cis106 error.log marstenHouse practicels vscode.sh
core.rb lab5files marstenhouse.sh Public
Desktop 'lab5files (2)' Music scripts
Documents 'lab5files (3)' 'New Document.txt' Templates
aryanvm@vbox:/usr$ ls -ltX ~/cis106
total 16
drwxrwxr-x 2 aryanvm aryanvm 4096 Oct 24 10:22 cheat_sheets
drwxrwxr-x 3 aryanvm aryanvm 4096 Oct 24 12:35 labs
drwxrwxr-x 2 aryanvm aryanvm 4096 Oct 24 10:22 notes
drwxrwxr-x 2 aryanvm aryanvm 4096 Oct 24 10:23 week_reports
aryanvm@vbox:/usr$ ls -Rr /home/aryanvm/cis106
/home/aryanvm/cis106:
week_reports notes labs cheat_sheets
/home/aryanvm/cis106/week_reports:
/home/aryanvm/cis106/notes:
/home/aryanvm/cis106/labs:
lab4
/home/aryanvm/cis106/labs/lab4:
challenge_lab4.sh.save challenge_lab4.sh
/home/aryanvm/cis106/cheat_sheets:
aryanvm@vbox:/usr$ ls -lh ~
total 3.5M
drwxrwxr-x 7 aryanvm aryanvm 4.0K Oct 24 10:40 cis-106
drwxrwxr-x 6 aryanvm aryanvm 4.0K Oct 24 10:23 cis106
15% 67% 29% ↑0.2 kB ↓0.2 kB Oct 28 11:30 PM


Dashboard - I
cis-106/notes
The Linux F
docs.google.com/presentation/d/e/2F
Listing Files and Directories (The ls command)
How to use it?
ls + option + directory to list
ls is used for listing the content of a given directory or the file/directory itself.
ls can be used with or without arguments.
Using ls without a file or directory lists the content of the present working directory.
ls has a lot of options. You can see them with: man ls
ls sorts the output alphabetically if no sorting option/arguments are given.
ls can sort the output:
By name (default)
By last modified
By file size
By extension
aryanvm@vbox: /usr
-rw-rw-r-- 1 aryanvm aryanvm 3.4K Oct 22 13:42 marstenhouse.sh
drwxr-xr-x 2 aryanvm aryanvm 4.0K Oct 23 10:22 Music
-rw-rw-r-- 1 aryanvm aryanvm 0 Oct 24 11:02 'New Document.txt'
drwxr-xr-x 3 aryanvm aryanvm 4.0K Oct 24 09:54 Pictures
drwxrwxr-x 2 aryanvm aryanvm 4.0K Feb 27 2022 practicels
drwxr-xr-x 2 aryanvm aryanvm 4.0K Oct 23 10:22 Public
drwxrwxr-x 2 aryanvm aryanvm 4.0K Oct 24 12:18 scripts
drwxr-xr-x 2 aryanvm aryanvm 4.0K Oct 23 10:22 Templates
drwxr-xr-x 2 aryanvm aryanvm 4.0K Oct 23 10:22 Videos
-rwxrwxr-x 1 aryanvm aryanvm 2.7K Oct 22 13:42 vscode.sh
aryanvm@vbox:/usr$ ls -lgo --full-time ~
total 3576
drwxrwxr-x 7 4096 2026-10-24 10:40:50.269267283 -0400 cis-106
drwxrwxr-x 6 4096 2026-10-24 10:23:11.973266779 -0400 cis106
-rwxrwxr-x 1 11292 2026-10-22 13:42:34.000000000 -0400 core.rb
drwxr-xr-x 2 4096 2026-10-28 23:10:15.850102966 -0400 Desktop
drwxr-xr-x 2 4096 2026-10-24 10:21:57.741266743 -0400 Documents
drwxr-xr-x 2 4096 2026-10-24 18:41:07.634575563 -0400 Downloads
-rw-rw-r-- 1 81 2026-10-24 08:52:30.091693140 -0400 error.log
drwxrwxr-x 2 4096 2022-02-22 14:18:06.000000000 -0500 lab5files
drwxrwxr-x 2 4096 2022-02-22 14:18:06.000000000 -0500 'lab5files (2)'
drwxrwxr-x 2 4096 2022-02-22 14:18:06.000000000 -0500 'lab5files (3)'
-rw-rw-r-- 1 3570561 2026-10-22 13:42:34.000000000 -0400 lab5files.zip
drwxrwxr-x 3 4096 2026-10-28 21:41:35.431058941 -0400 marstenHouse
-rw-rw-r-- 1 3407 2026-10-22 13:42:34.000000000 -0400 marstenhouse.sh
drwxr-xr-x 2 4096 2026-10-23 10:22:41.424000006 -0400 Music
-rw-rw-r-- 1 0 2026-10-24 11:02:18.481267898 -0400 'New Document.txt'
drwxr-xr-x 3 4096 2026-10-24 09:54:50.601265967 -0400 Pictures
drwxrwxr-x 2 4096 2022-02-27 13:46:18.000000000 -0500 practicels
drwxr-xr-x 2 4096 2026-10-23 10:22:41.424000006 -0400 Public
drwxrwxr-x 2 4096 2026-10-24 12:18:29.221270077 -0400 scripts
drwxr-xr-x 2 4096 2026-10-23 10:22:41.424000006 -0400 Templates
drwxr-xr-x 2 4096 2026-10-23 10:22:41.424000006 -0400 Videos
-rwxrwxr-x 1 2681 2026-10-22 13:42:34.000000000 -0400 vscode.sh
aryanvm@vbox:/usr$ ls -lgo --full-time ~
total 3576
drwxrwxr-x 7 4096 2026-10-24 10:40:50.269267283 -0400 cis-106
drwxrwxr-x 6 4096 2026-10-24 10:23:11.973266779 -0400 cis106
30% 65% 29% ↑0.2 kB ↓0.2 kB Oct 28 11:30 PM

scipts


Dashboard -
cis-106/notes
Shell Scripting
docs.google.com/presentation/d/e/2PACX-
Practice 2: Using environment variables
1. Create a shell script called: script_env_var.sh
2. Save the script in the "Scripts" directory located in your home directory. You should have this directory already
don't, create it.
3. Open a terminal and type the env command to list all the environment variables.
4. In your script use the echo command to display the following:
a. The current user
b. The current shell
c. The current user's home directory absolute path
d. The path variable (this is an important variable! read here.)
5. Save and execute the script
aryanvm@vbox: ~/Scripts
UPTIME: up 0 minutes
aryanvm@vbox:~$ cd ~/Scripts
aryanvm@vbox:~/Scripts$ ls
aryanvm@vbox:~/Scripts$ find ~ -name "script_env_var.sh"
/home/aryanvm/cis-106/labs/lab4/script_env_var.sh
/home/aryanvm/script_env_var.sh
aryanvm@vbox:~/Scripts$ cp ~/script_env_var.sh ~/Scripts/
aryanvm@vbox:~/Scripts$ ls
script_env_var.sh
aryanvm@vbox:~/Scripts$ bash script_env_var.sh
Current User: aryanvm
Current Shell: /bin/bash
Home Directory: /home/aryanvm
PATH: /usr/local/bin:/usr/bin:/bin:/usr/local/games:/usr/games:/snap/bin:/usr/bi
n
aryanvm@vbox:~/Scripts$ chmod +x script_env_var.sh
aryanvm@vbox:~/Scripts$ ./script_env_var.sh
Current User: aryanvm
Current Shell: /bin/bash
Home Directory: /home/aryanvm
PATH: /usr/local/bin:/usr/bin:/bin:/usr/local/games:/usr/games:/snap/bin:/usr/bi
n
aryanvm@vbox:~/Scripts$
Search history
View
Today
Ask ChatGPT
High
12% 65% 0% 0.2 kB 0.3 kB Oct 28 11:56 PM


Open
script_usr_var.sh
~/scripts
lab4.script2.sh
lab4.script1.sh
lab4.script3.sh
challenge_lab4.sh
example.sh
lab 5 notes.txt
script_env_var.sh
script_usr_var.
#!/bin/bash
name="bob"
age=47
email="bob@example.com"
phone="123-333-4477"
echo "User info"
echo "Name: $name"
echo "Age: $age"
echo "Email address: $email"
echo "Phone number: $phone"
aryanvm@vbox: ~/scripts
DATE OF INSTALL: N/A
DATE OF LAST UPDATE: 2026-10-24 18:08:36
TOTAL PACKAGES: 1855
UPTIME: up 19 minutes
aryanvm@vbox:~$ cd ~/Scripts
bash: cd: /home/aryanvm/Scripts: No such file or directory
aryanvm@vbox:~$ cd ~/scripts
aryanvm@vbox:~/scripts$ ls
challenge_lab4.sh lab4.script2.sh script_env_var.sh
lab4.script1.sh lab4.script3.sh script_usr_var.sh.
aryanvm@vbox:~/scripts$ mv script_usr_var.sh. script_usr_var.sh
aryanvm@vbox:~/scripts$ bash script_usr_var.sh
User info
Name: bob
Age: 47
Email address: bob@example.com
Phone number: 123-333-4477
aryanvm@vbox:~/scripts$ chmod +x script_usr_var.sh
aryanvm@vbox:~/scripts$ ./script_usr_var.sh
User info
Name: bob
Age: 47
Email address: bob@example.com
Phone number: 123-333-4477
aryanvm@vbox:~/scripts$
3% 72% 0% 0.0 kB 0.0 kB Oct 29 12:14 AM